PHASE 3 — Basic Retrieval-Augmented Generation

Objective

Implement a basic RAG pipeline that answers questions using the local pharmacology corpus.

This phase must produce a working question-answering flow before agentic orchestration is introduced.

Expected workflow

User question
→ Query embedding
→ Semantic retrieval
→ Relevant context construction
→ LLM answer generation
→ Answer with source citations

Step 1 — LLM integration

Use the configured chat model:

- OPENAI_API_KEY
- OPENAI_BASE_URL
- OPENAI_MODEL

Use the existing OpenAI-compatible SDK configuration.

Verify that the chat model is available.

Do not assume that the chat model is identical to the embedding model.

Step 2 — Context construction

Use the retrieval subsystem from Phase 2.

Construct a context containing:

- Retrieved passages.
- Source filenames.
- Section metadata.

- Chunk identifiers where useful.

Avoid sending the entire corpus to the LLM.

Implement a reasonable context-size limit.

Do not silently remove source metadata.

Step 3 — Answer generation

Create a generation component.

The model must:

- Answer using the provided evidence.
- Avoid inventing unsupported information.
- Identify insufficient evidence.
- Preserve important qualifications.
- Include source citations.
- Distinguish source-supported findings from uncertainty.

Include the medical-information disclaimer.

Step 4 — Citation handling

Citations must originate from retrieved source metadata.

Do not ask the model to invent citation filenames.

Keep source references available to the application so citations can be validated.

Step 5 — Tests

Test:

- Normal questions.
- Questions with no relevant results.
- Questions with insufficient evidence.
- Multiple retrieved sources.
- Citation generation.
- LLM failures.
- Context-size limitations.

Mock the LLM in unit tests.

Constraints

- Do not implement planning.
- Do not implement agent tool calling.
- Do not implement dynamic tool selection.
- Do not implement the critic yet.
- Do not implement the frontend.

Deliverables

Provide:

1. A description of the RAG pipeline.
2. An example request and response.
3. Explanation of grounding and citations.
4. Test results.
5. Known limitations.